Intel's latest Xeon D-2100 processor chip helps customers who want to move computing to the edge. It's part of an effort by the chip giant to stay ahead of emerging technology trends like edge computing and the Internet of Things.

Computing at the edge has some unique space and power requirements that Intel has tried to address with this announcement. For starters, it provides a standalone system-on-a-chip (SoC). This means everything you need (compute, networking and storage) is built into the chip. It's also low-power, which might be necessary to run an edge computing device without the benefit of a data centre power structure.

The SoC gives customers hardware-enhanced network, security and acceleration capabilities in a single package. The chip achieves all of this by packing up to 18 Skylake-server generation Intel Xeon processor cores integrated with up to 100Gbps of built-in cryptography, decryption and encryption acceleration. Intel calls this QuickAssist technology.

The company considers it particularly useful for new 5G technologies being developed for smartphones, including augmented reality, virtual reality and autonomous driving.

The era of quantum supremacy arriving

After decades of not much success, quantum computing is suddenly buzzing with activity. IBM and Intel have made quantum computers with 50 and 49 qubits, respectively, and Google is also thought to have its own in the wings.

In fact, there is now talk of quantum supremacy—the moment when a quantum computer can carry out a task beyond the means of today's best classical supercomputers. That might sound absurd when you compare the numbers: 50 qubits versus the billions of classical bits in your laptop. But the whole point of quantum computing is that, a quantum bit counts for much, much more than a classical bit. Fifty qubits has long been considered the approximate number at which quantum computing becomes capable of calculations that would take an unfeasibly long time classically.

IBM's quantum computing centre at the Thomas J. Watson Research Center in Yorktown Heights, New York, holds quantum computers in large cryogenic tanks (far right) that are cooled to a fraction of a degree above absolute zero

Both the benefits and the challenges of quantum computing are inherent in the physics that permits it. Classical computers encode and manipulate information as strings of binary digits—1 or 0. Quantum bits do the same, except that they may be placed in a so-called superposition of the states 1 and 0, which means that a measurement of the qubit's state could elicit the answer 1 or 0 with some well-defined probability.

To perform a computation with many such qubits, they must all be sustained in interdependent superpositions of states—a quantum-coherent state, in which the qubits are said to be entangled. That way, a tweak to one qubit may influence all the others. This means that somehow computational operations on qubits count for more than they do for classical bits. The computational resources increase in simple proportion to the number of bits for a classical device, but adding an extra qubit potentially doubles the resources of a quantum computer.

Police in China using facial-recognition glasses

Police in China are now sporting glasses equipped with facial recognition devices to scan train and plane passengers for individuals trying to evade law enforcement or using fake IDs. So far, they have caught seven people connected to major criminal cases and 26 with false IDs.

Police in China have been equipped with facial-recognition glasses to look for criminals and travellers using fake IDs (Photo credit: AFP/Getty Images)

The *Wall Street Journal* reports that Beijing-based LLvision technology company developed the devices. The company produces wearable video cameras as well and while it sells those to anyone, it vets buyers for its facial recognition devices.

LLvision says that in tests, the system was able to screen out individuals from a database of 10,000 people and it could do so in 100 milliseconds. However, in the real world, accuracy would probably drop due to environmental noise. Additionally, aside from portability, another difference between these devices and typical facial recognition systems is that the database used for comparing images is contained in a handheld device rather than the cloud.

Obviously, there are privacy concerns regarding this technology and not everyone believes police should be using it.